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The Big Read **US**

Why America is struggling to power the AI boom

The US is building electricity lines at a quarter the rate of a decade ago

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It took 13 years to build a 102-mile transmission line to carry electricity from Iowa to Wisconsin — more than a decade of delays and blown budgets that signal trouble ahead for America's AI ambitions.

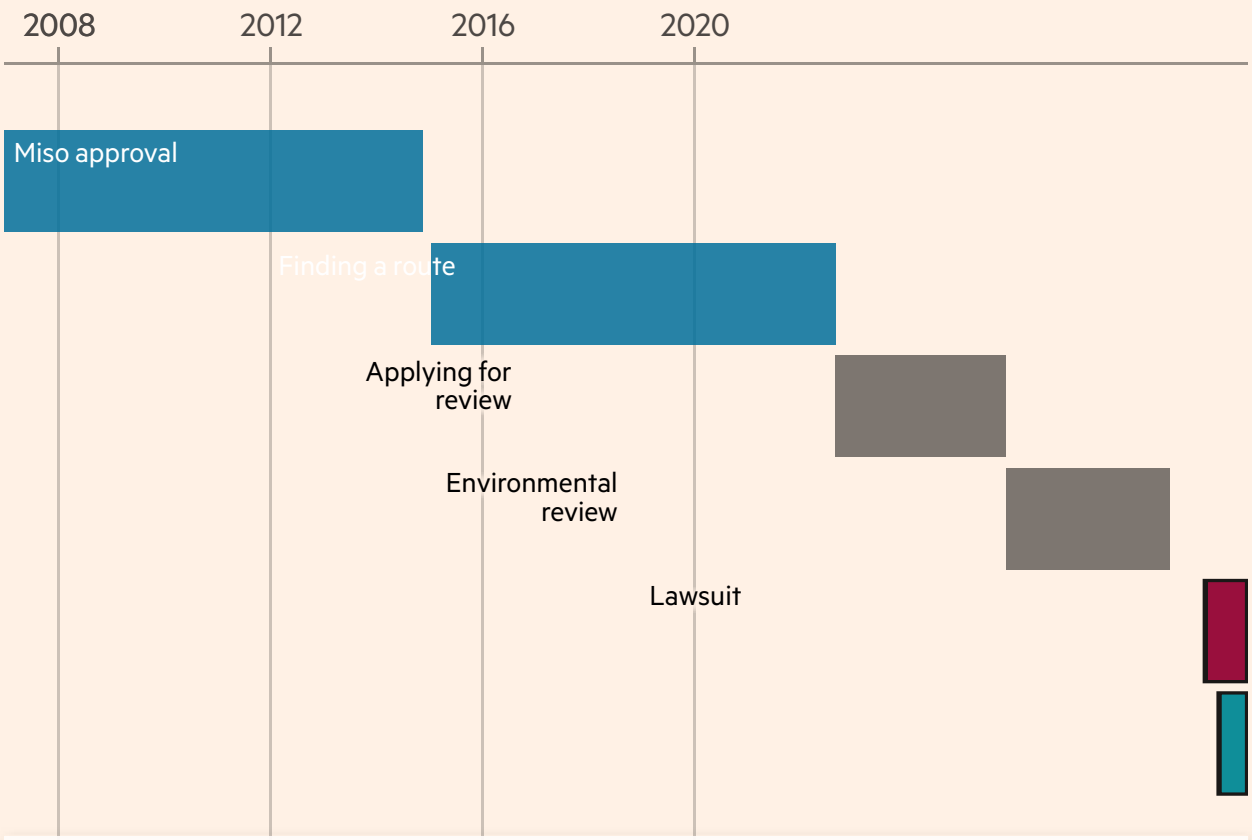
The Cardinal-Hickory Creek line takes power from a wind farm near the tiny town of New Vienna in Iowa, across the Mississippi River and through a national wildlife and fish refuge, to the affluent commuter hub of Middleton in neighbouring Wisconsin.

The line was intended to relieve congestion in the US electricity transmission grid — a goal that has rarely, if ever, been so important.

America's battle with China for global supremacy in AI will require unprecedented expenditure on data centres, which in turn will necessitate a vast, co-ordinated expansion of not just the nation's power stations but of the wires and poles that carry electricity across the country.

"We're in a global race," says Krista Tanner, chief executive of ITC, one of the groups behind the Cardinal-Hickory Creek project. "Whether it's AI or advanced manufacturing," she adds, the common question is: 'Do we have the infrastructure in place to build it?' Increasingly, I worry that the answer is no."

As the tale of the connection across the Mississippi shows, the same combination of federal and local systems that historically fostered American competition, innovation and growth may now be a liability when it comes to building infrastructure on a national scale and at speed.



In December 2011, after a three-year review process, the Midcontinent Independent System Operator (Miso) approves the need for Cardinal-Hickory Creek, but not its construction.

Finding a route takes years. Developers send 23,474 invitations to landowners and officials for public meetings in three Iowa and Wisconsin communities.

Federal environmental review begins. Wisconsin twice declares the developer's application incomplete, demanding corrections to route maps, environmental information, costs and economic modelling.

Wisconsin approves the line in September 2019, followed by key federal permissions in September 2020. Nearly nine years after Miso's decision, the project appears ready to proceed.

Conservation groups sue in February 2021. Construction begins in April, but in August the US Fish and Wildlife Service revokes a permit allowing the line to cross the Upper Mississippi River wildlife refuge, leaving a crucial link unfinished.

A federal judge voids the environmental review and federal project approval. Developers seek a workaround: a land swap with the wildlife refuge for the missing crossing. Work continues elsewhere.

With 101 of 102 miles built, the line still cannot operate in full. A judge blocks the land swap in March 2024 but then an appeals court clears it in May, allowing work to begin on the final 1.1 miles.

Finally, in September 2024, electricity flows along the full line – nearly 13 years after Miso's approval and almost 40 per cent over its 2019 budget.

Tanner says that when she joined ITC in 2014, the transmission line was the subject of her very first meeting. “I thought we’d build it pretty quickly,” she recalls.

But what followed was a tortuous approval and construction process. The project was only completed in 2024, delayed not just by fights over its proposed path through the wildlife refuge but also by incompatibilities between Iowa, Wisconsin and federal rules.

Critics say such bottlenecks have held up construction of new US infrastructure for decades. Now, with the AI rollout at the heart of both the country's growth model and its geopolitical ambitions, they threaten to become more of a constraint than ever before.

Electricity is to AI what coal was to the Industrial Revolution.

So-called hyperscalers such as Amazon, Google, Meta, Microsoft and Oracle are expected to spend around $900bn on AI-related projects this year, according to JPMorgan estimates. Their ambitions depend on access to vast amounts of electricity carried over an ageing grid that the US is struggling to revamp and expand.

After two decades in which energy usage barely grew, AI is driving a sudden and sharp rise in demand. Grid Strategies, a consultancy, says demand will increase by 166 gigawatts by 2030 — equivalent to 15 New York Cities.



From new nuclear reactors to ageing coal-fired plants hoping for a revival, the race to build new capacity has also attracted enormous investment. Generating more electricity is only half the battle; it must also reach the places that need it.

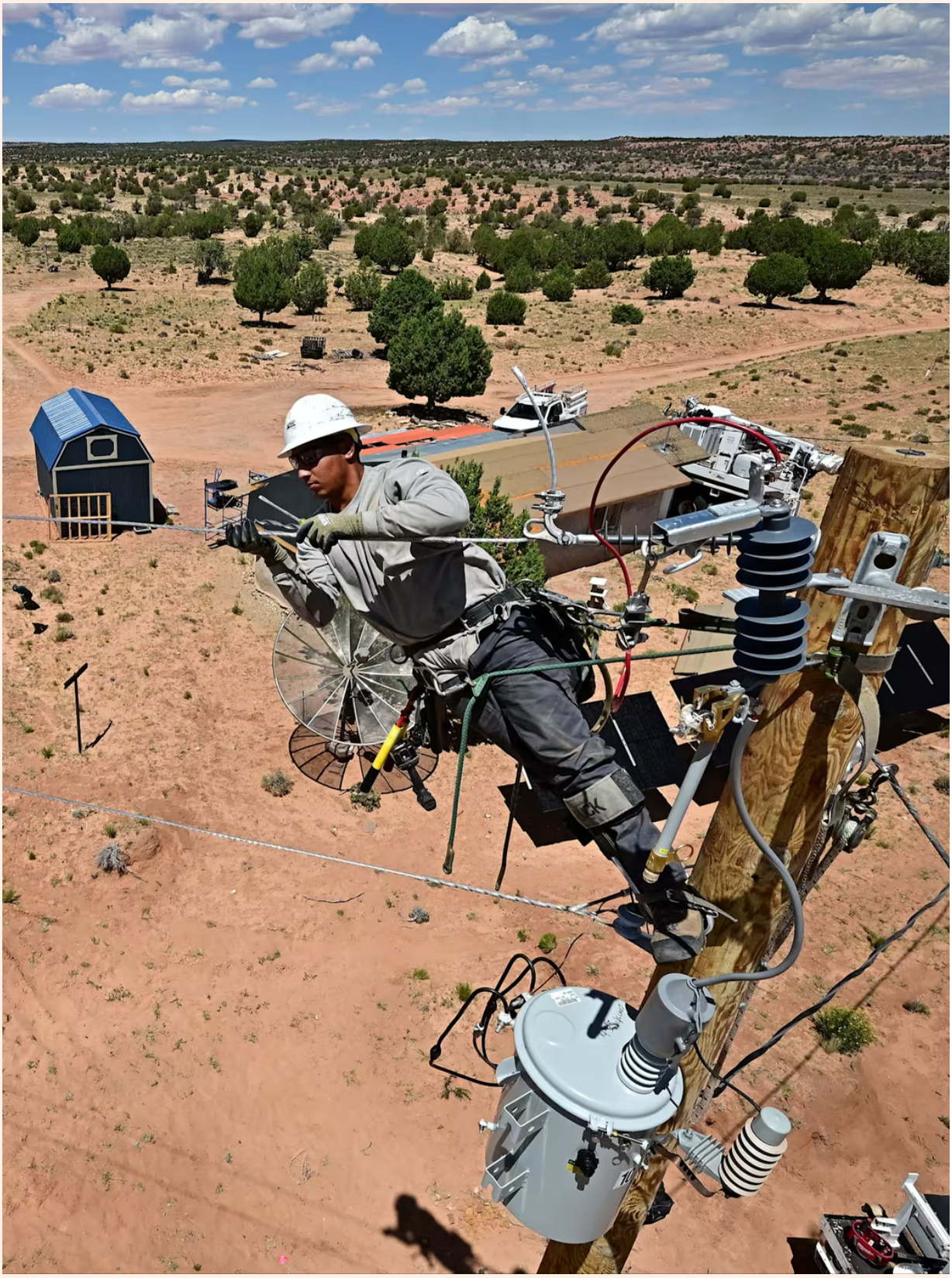
Transmission lets electricity from regions with cheap, abundant supplies reach areas where demand is greatest, relieving bottlenecks and lowering costs. It also allows regions to share power during shortages, reducing the risk of blackouts.

But much of the country's transmission network, built during the construction boom of the 1960s and 1970s, is now approaching the end of its expected lifespan.

Development of new high-voltage power lines — which carry electricity over long distances — has slowed dramatically. The US built nearly 4,000 miles of transmission in 2013, but by 2024 the rate had slumped to less than 900 miles. Major transmission projects take on average a decade to develop, twice as long as it took to build the Hoover Dam in the 1930s.

Infrastructure bottlenecks are not confined to the US. The International Energy Agency says that new transmission lines also take between five and 15 years to plan and build in Europe, while delays are also commonplace in Canada, Australia and New Zealand.

What sets the US apart is the extent to which hold-ups are caused by overlapping authorities, with no single institution able to plan and execute major inter-regional projects. States excel at drafting copious regulation but struggle to build, an argument set out at length in Ezra Klein and Derek Thompson's book *Abundance*.



America’s battle for global supremacy in AI will necessitate a vast, co-ordinated expansion of the nation’s power infrastructure © Frederic J Brown/AFP/Getty Images



Sections of metal electrical towers waiting to be installed near data centres in Washington state. The country's transmission grid has evolved in piecemeal fashion © Scott Brauer/ZUMA/Reuters

The contrast with China, which has built more transmission lines in the past 15 years than the US has in its entire history, is vivid. Beijing can centrally plan and direct infrastructure development, while US projects must navigate a fractured system of utilities, landowners, grid operators and state and federal regulators.

Americans who object to a new development can turn to the courts or petition local politicians, whereas Chinese citizens have far less scope to oppose projects.

“We’ve had over a century of being able to power industries that drive growth,” says Ari Peskoe, director of the Electricity Law Initiative at Harvard Law School. “If we can’t do that anymore, we’ll lose our economic vitality.”

Edison and beyond

The first US power transmission line opened in Oregon in 1889, a decade after Thomas Edison invented the lightbulb. From then, America’s transmission grid evolved in piecemeal fashion — entrepreneurs and communities formed private companies that built power plants, erected poles and wires and connected up local customers.

Such regional entities initially resisted government oversight but eventually embraced a degree of regulation in exchange for monopoly franchises that shielded them from localised competition.

The system still lacked a guiding authority, taking a longer-term view and deciding what and how to build.

“There has never been an electricity tsar in the federal government or a single entrepreneur whose company dominated the whole industry,” says Julie Cohn, a non-resident scholar at Rice University’s Center for Energy Studies. “So there’s never been a single game plan that everybody agreed to.”

Decentralisation has traditionally been a source of US economic strength. With powers dispersed among states, cities, businesses and civic institutions, local leaders could experiment with solutions tailored to their regions. Population growth and new jobs encouraged innovation and investment, with successful ideas often spreading across the country.



High-voltage power lines run through a substation in Florida. US projects must navigate utilities, landowners, grid operators and state and federal regulators © Joe Raedle/Getty Images



The transformer substation for China's 500kV ultra-high-voltage power line in Zhejiang province. Beijing can centrally plan and direct infrastructure development © Imagine China/Reuters

That becomes a weakness when projects start to cross local and state boundaries. What might work for an individual city is harder to replicate in a second, or in a city in another state.

“Someone, somewhere is always innovating on something. There is genius in that,” says Bruce Katz, founder of consulting firm New Localism Associates. “But we have to focus on how to spread and how to scale; it just can’t be complete chaos.”

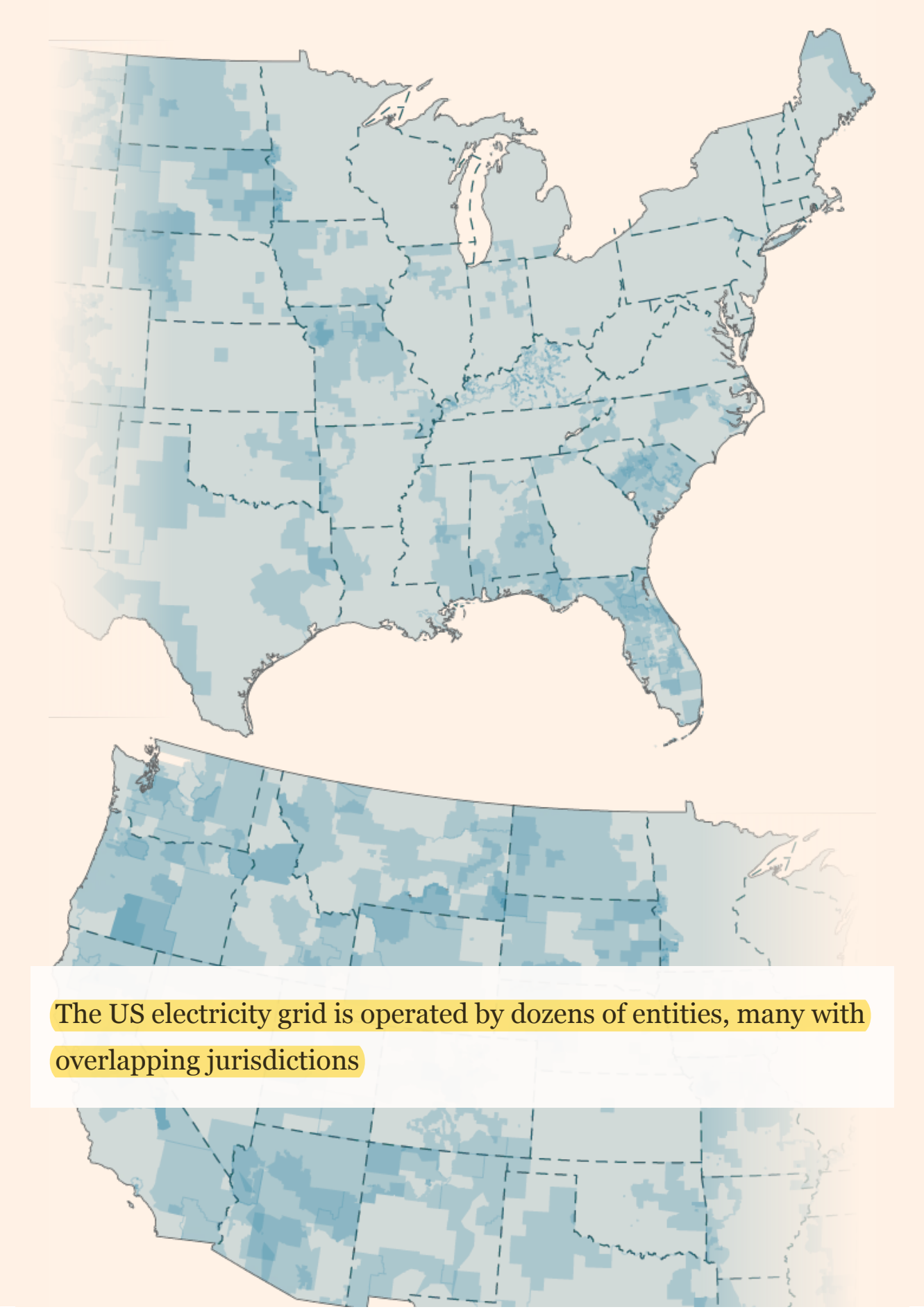
The US electricity system clearly shows the trade-off. At the bottom layer, utilities and independent power producers own and develop infrastructure. Regional grid operators such as Midcontinent Independent System Operator (Miso), which covers the Midwest and parts of the South, co-ordinate electricity flows and identify transmission needs.

State regulation influences what utilities can build and charge, while the Federal Energy Regulatory Commission sets rules for regional planners.

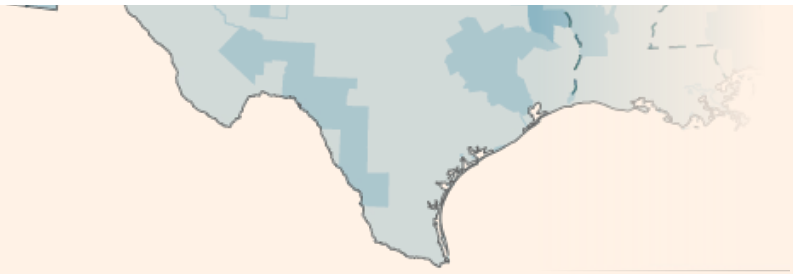
Such bodies’ responsibilities and incentives do not always align, leading to disagreements over whether a line is needed and who should pay for it.

Utilities often resist long-distance lines that could open up their local catchment areas to potentially cheaper power from elsewhere. The seven grid operators rarely plan large inter-regional transmission projects together. States with green energy mandates might object to transmission lines that bring coal-generated power into their areas.

Grid operators can also operate in more than a dozen states each, often complicating projects that cross state borders.



The US electricity grid is operated by dozens of entities, many with overlapping jurisdictions



Larger regional transmission organisations (RTOs) and independent system operators (ISOs) are federally regulated, independent non-profits that control and monitor multi-state electric power transmission grids.

Electricity transmission lines criss-cross the country, often traversing the jurisdictions of competing operators

“Normally we think our political boundaries matter,” says Chaz Teplin, a principal in think-tank RMI’s carbon-free electricity programme. “But our grid boundaries are mismatched and mixed up. So who gets to decide? Lots of people.”

Even when planners reach agreement, there is no guarantee lines will be built. Federal regulators require grid operators like Miso to identify future infrastructure needs, but those operators cannot compel companies such as ITC to build lines.

“You can mandate all of this thinking about what lines need to be built. But at no point does anyone have to build anything,” says Advait Arun, senior associate for capital markets at the Center for Public Enterprise, a think-tank.

Once projects are approved, they have to secure an array of different permissions and permits from landowners, states and federal agencies.

Developers must negotiate with individual landowners for the right to build across their properties, with every stretch of transmission line bringing another negotiation.

“I’ve sat around kitchen tables with landowners and met their kids and grandkids,” says Casey Carroll, senior vice-president for energy infrastructure development at LS Power. “But there are also landowners who just want it to be somewhere else.”

No federal statute looms larger over grid development than the National Environmental Policy Act. Passed in 1970, it requires federal agencies to assess projects’ impact on wildlife, waterways, habitats and communities. The process can involve years of studies and public consultation.





The Cardinal-Hickory Creek transmission line from Iowa to Wisconsin went through a tortuous approval and construction process, taking 13 years to complete © ITC





Some developers are exploring 'behind the meter' power plants to supply facilities. AWS built a data centre next to the Susquehanna nuclear power plant in Pennsylvania © Ted Shaffrey/AP

Even after permission is granted, interest groups can challenge the decisions in court — as they did in the case of Cardinal-Hickory Creek — potentially leaving developers on the hook for years of uncertainty.

"Permitting can take two to four years," says Ben Porath, executive vice-president and chief operating officer at Dairyland Power Cooperative, one of the project's co-owners. "But it's the multiple years in the court system that really bogs these projects down."

Of 18 lawsuits researched by the Niskanen Center, a think-tank, 16 ultimately failed. But cumulatively they add decades of delay; more than a quarter of major US transmission projects it analysed undergoing federal review between 2010 and 2020 were delayed or cancelled due to opposition and litigation, it found.

"It's an incredible risk, so why would you invest?" says Rob Gramlich, the president of Grid Strategies, a consultancy. "Everybody wants clean water and clean air, but we have to speed up the process and limit never-ending litigation exposure."

'The system is getting tighter and tighter'

Industry experts warn that the US's failure to expand the electricity grid risks strangling the AI boom and handing an advantage to China, which can plan grid infrastructure centrally and can build it much more quickly.

“If energy becomes a bottleneck, it will become harder and harder to build AI,” says Dana Golden, an economist specialising in energy infrastructure.

Grid gaps also threaten to drive up electricity costs. Without sufficient transmission lines, cheap electricity generated in one region cannot reach customers in others, leaving them dependent on potentially more expensive power from a local supplier.

“We already have much more expensive electricity than China, because we’re not building,” says Zachary Liscow, a professor at Yale Law School. “It’s a burden on consumers.”

While deep-pocketed hyperscalers can afford to pay a premium for power, that could squeeze out other industrial users operating on thinner margins and threaten President Donald Trump’s agenda of reviving manufacturing.

“There’s a lot of on-the-margin factories that barely make money. If their prices go up, they might have to either dial back a few hours a day, relocate or shut down,” says Michael Webber, a professor of mechanical engineering at the University of Texas at Austin.

Fears that data centres will push up local electricity costs by increasing demand on a stressed system have also catalysed local opposition, with half of Americans polled by Pew Research Center saying data centres are bad for household energy costs.





A protest in Las Vegas in June. Half of Americans polled by Pew Research Center say data centres are bad for household energy costs © Bryan Steffy/Getty Images for People's Action



There are concerns that an AI boom could cause enough strain on the grid to require electricity rationing © Justin Sullivan/Getty Images

Utilities are planning vast investments in power plants and transmission lines — $1.3tn by 2030, according to S&P — much of which will ultimately be financed by customers through their utility bills. Even when technology companies agree to fund upgrades themselves, high demand for gas turbines, electrical equipment and workers will cause knock-on effects across the economy.

Data centre operators argue that they could help bring electricity rates down in the long run by helping utilities spread the fixed costs of infrastructure upgrades.

How AI will be monetised in future is still vigorously debated, meaning utilities face the risk of spending billions on infrastructure for data centres that are delayed or abandoned because demand for their services was overestimated. “Ratepayers could be left footing the bill,” warns Jesse Buchsbaum, an economist at Resources for the Future, a research nonprofit.

Alternatively, an AI boom could cause enough strain on the grid to require rationing, with data centres and even smaller customers asked to curtail operations during periods of peak demand. In the bleakest scenario, blackouts are a possibility.

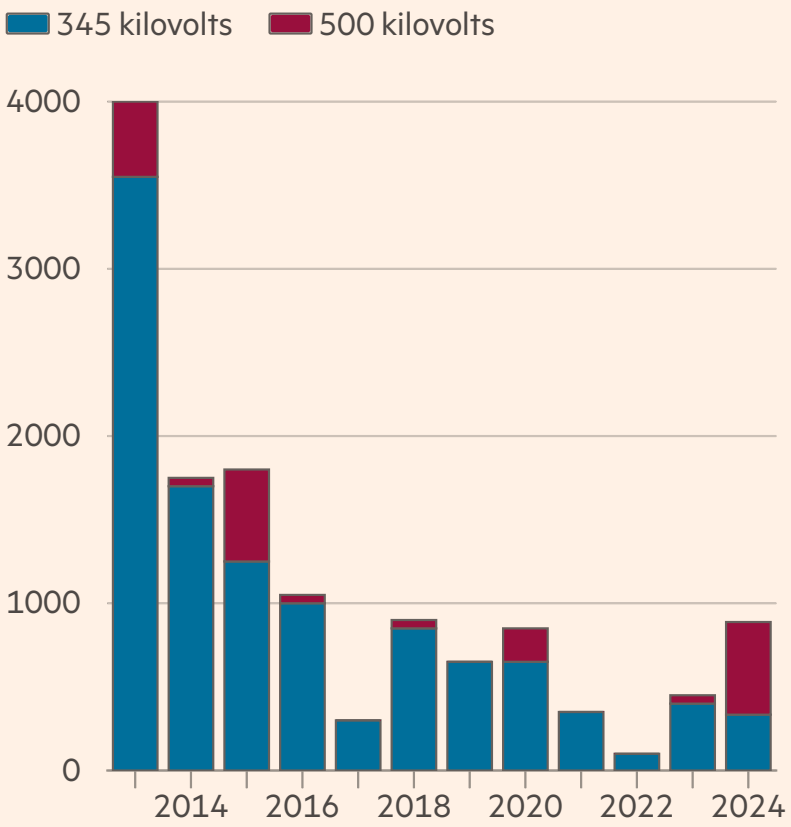
“The system is getting tighter and tighter,” says Joseph Bowring, president of Monitoring Analytics, part of grid operator PJM. “Hopefully it’s managed curtailments, because blackouts are a disaster.”

America’s lawmakers know the grid is in bad shape. In September a group of senators introduced the Bipartisan American Affordability and Jobs Act, to accelerate federal reviews for transmission and impose tighter deadlines on permits and lawsuits.

But some say that the focus on new power plants and transmission lines is misplaced when the US could squeeze more out of the grid it already has.

The US is building far less grid capacity than it used to

New transmission lines, miles



Source: Ferc

Financial Times

Advanced conductors could replace older wires, while sensors that monitor weather and line temperatures could allow operators to adjust how much power they can safely carry.

The grid's biggest customers could also learn to take turns. So-called demand flexibility would allow power-hungry data centres to delay non-essential tasks, shift workloads to other locations or use backup power during periods of strain, freeing up capacity for other users. In March, Google signed agreements for its data centres in Tennessee, Arkansas and Michigan that allow utilities to ask it to reduce power usage when the grid is under strain, in return for faster grid access.

Jigar Shah, an energy investor and former Department of Energy official, says all of the US's electricity needs for the next decade could be met through a combination of demand flexibility and modest infrastructure upgrades.

"Every single utility in the country is going to move towards flexibility because there's no other way to do it," he adds.

But utilities, whose regulation is generally based on a return on capital invested in infrastructure, have a strong incentive to build new assets rather than squeeze more out of existing ones. They also dispute that flexibility alone will be enough to meet the surge in demand.

"These tools have their place, but they don't provide that level of scale," says David Stader, chief transmission development officer at American Electric Power. "This is not a simple matter."

Impatient data centre developers are increasingly taking matters into their own hands by building their own "behind the meter" power plants to supply their facilities. Research firm Cleanview says 59 sites, representing over a quarter of all planned US data centre capacity, are planning on-site power. But going it alone is expensive, with developers facing shortages of equipment and the need to build backup capacity to remain online if their plants break down or require maintenance.

Ben Alingh, the co-founder of site developer Monarch Energy, worries that the US's data centre developers do not have time for a protracted debate about fixing energy bottlenecks. For years, companies such as Monarch have hunted for pockets of spare power capacity where data centres could be built without upgrades to energy infrastructure.

Those havens are fast disappearing. The 10 years it now takes to build a new transmission line is an eternity for an industry that is moving at breakneck speed.

"The data centre boom is predicated on access to electricity," he warns. "Ten years out is essentially never."

Data visualisation by Max Köhler. Cartography by Steven Bernard and Kripa Pancholi

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